Functions

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Abstract

A function takes something in and gives something out. It does not explain, justify, or remember. It transforms. The function $\zeta(s)$ takes a complex number in and returns a complex number out. It says nothing about why. Its zeros are the places where the output is zero: the inputs that the function annihilates. RH is the question: where are the annihilation points? Are they arranged? The function does not know. We must read it.

1 What a Function Is

Definition 1 (Function)**.** A function $f : X \to Y$ is a rule assigning to each element of X exactly one element of Y. It says nothing beyond the assignment. It has no history, no motive, no preference. It returns the same output for the same input, always.

Remark 2 (Functions do not explain). The function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ assigns to each s a sum. It does not explain why the sum converges for $\mathrm{Re}(s) > 1$, why it extends to the whole plane, why it has zeros at $-2, -4, -6, \ldots$ (the trivial zeros), why the nontrivial zeros appear to cluster on $\mathrm{Re}(s) = 1/2$. It just does. The function is the record, not the reason. Reading the function means inferring the reason from the record. RH is the question: is there a pattern in the record?

2 The Functions in This Body

Remark 3 (The functions that appear). This body contains many functions. Each paper is a function: it takes questions in and returns partial answers. The BN functions $\rho_{1/k}(x) = \{1/(kx)\} - (1/k)\{1/x\}$ are functions. The Gram matrix G_N is a function (of N, of the integers $2, \ldots, N+1$). The distance d_N is a function of N. The zero-counting function $N(\sigma, T)$ is a function of two variables. The density spectrum $\Lambda(\sigma)$ is a function of σ.

Each of these functions says something about the integers. Each says it without explanation. Reading them together is the project of this body.

Definition 4 (The characteristic function)**.** The function $1_{(0,1)} : \mathbb{R} \to \{0, 1\}$ is the simplest. It assigns 1 to $(0, 1)$ and 0 to everything else. It has no complexity. It makes no choices. It simply marks a region. All of this work is an attempt to reach this function from the integers alone. The simplest function is the hardest target.

3 Functions of Time

Remark 5 (The BN flow as a function of time). The flow $N \mapsto d_N^2$ is a function of the step N (which we may think of as time). It is decreasing: $d_{N+1}^2 \leq d_N^2$. It is positive: $d_N^2 > 0$ for all finite N. It converges: $d_N^2 \to d_\infty^2 \geq 0$. $d_\infty^2 = 0$ if and only if RH.

The function of time $N \mapsto d_N^2$ encodes RH in its limit. A function that converges to 0 proves RH. A function that stabilizes above 0 disproves it. RH is a question about the long-run behavior of a function of time.

Theorem 6 (The function that proves RH)**.** *To prove RH, construct explicitly a function $N \mapsto f_N \in \mathcal{V}_N$ such that $\|1_{(0,1)} - f_N\|_{L^2} \to 0$. Such a function would witness $d_N \to 0$ directly. No such function is currently known that does not assume RH. The function exists (under RH) but we cannot write it down. This is the situation: we know the function exists; we do not know the function.*

4 The Function and the Human

Remark 7 (What functions cannot do). Functions cannot intend. They cannot want. They return what they return regardless of preference. This is their power: they are reliable. This is their limit: they cannot surprise themselves.

A human is not a function. A human can intend, want, be surprised. A human can decide to pursue a question without knowing the answer. The Riemann Hypothesis is a question a function cannot ask. The function ζ does not know it has zeros. The human who studies ζ knows, and wants to understand.

The gap between function and human is the gap between description and understanding. Functions describe. Humans understand (or try to). This body is the attempt.

References

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